corresponding methods without at least the continuity of the representation. Stability is thus important but maybe a too strong requirement for many problems in data sciences. Designing linear representation that are sensitive to specific parts of a persistence diagrams rather than globally stable may reveal a good strategy in practice.

6 Statistical aspects of persistent homology

Persistence homology by itself does not take into account the random nature of data and the intrinsic variability of the topological quantity they infer. We now present a statistical approach to persistent homology, in the sense that data is considered as generated from an unknown distribution. We start with several consistency results for persistent homology inference.

6.1 Consistency results for persistent homology

Assume that we observe n points $(X_1, \ldots, X_n)$ in a metric space (M, ρ) drawn i.i.d. from an unknown probability measure μ whose support is a compact set denoted $\mathbb{X}_\mu$. The Gromov-Hausdorff distance allows us to compare $\mathbb{X}_\mu$ with compact metric spaces not necessarily embedded in M. In the following, an *estimator* $\widehat{\mathbb{X}}$ of $\mathbb{X}_\mu$ is a function of $X_1 \ldots, X_n$ that takes values in the set of compact metric spaces.

Let $\mathrm{Filt}(\mathbb{X}_\mu)$ and $\mathrm{Filt}(\widehat{\mathbb{X}})$ be two filtrations defined on $\mathbb{X}_\mu$ and $\widehat{\mathbb{X}}$. Starting from Theorem 9; an natural strategy for estimating the persistent homology of $\mathrm{Filt}(\mathbb{X}_\mu)$ consists in estimating the support $\mathbb{X}_\mu$. Note that in some cases the space M can be unknown and the observations $X_1 \ldots, X_n$ are then only known through their pairwise distances $\rho(X_i, X_j)$, $i, j = 1, \cdots, n$. The use of the Gromov-Hausdorff distance allows us to consider this set of observations as an abstract metric space of cardinality n, independently of the way it is embedded in M. This general framework includes the more standard approach consisting in estimating the support with respect to the Hausdorff distance by restraining the values of $\widehat{\mathbb{X}}$ to the compact sets included in M.

The finite set $\mathbb{X}_n := \{X_1, \ldots, X_n\}$ is a natural estimator of the support $\mathbb{X}_\mu$. In several contexts discussed in the following, $\mathbb{X}_n$ shows optimal rates of convergence to $\mathbb{X}_\mu$ with respect to the Hausdorff distance. For some constants $a, b > 0$, we say that μ satisfies the (a, b)-standard assumption if for any $x \in \mathbb{X}_\mu$ and any $r > 0$,

$$\mu(B(x, r)) \geq \min(ar^b, 1). \tag{6}$$

This assumption has been widely used in the literature of set estimation under Hausdorff distance (Cuevas and Rodríguez-Casal, 2004; Singh et al., 2009). Under this assumption, it can be easily derived that the rate of convergence of $\mathrm{dgm}(\mathrm{Filt}(\mathbb{X}_n))$ to $\mathrm{dgm}(\mathrm{Filt}(\mathbb{X}_\mu))$ for the bottleneck metric is upper bounded by $O\left(\frac{\log n}{n}\right)^{1/b}$. More precisely, this rate upper bounds the minimax rate of convergence over the set of probability measures on the metric space (M, ρ) satisfying the (a, b)-standard assumption on M.

Theorem 10. [Chazal et al. (2014b)] *For some positive constants a and b, let*

$$\mathcal{P} := \left\{\mu \text{ on } M \mid \mathbb{X}_\mu \text{ is compact and } \forall x \in \mathbb{X}_\mu, \forall r > 0,\ \mu\left(B(x, r)\right) \geq \min\left(1, ar^b\right)\right\}.$$

Then, it holds

$$\sup_{\mu \in \mathcal{P}} \mathbb{E}\left[\mathrm{d_b}(\mathrm{dgm}(\mathrm{Filt}(\mathbb{X}_\mu)), \mathrm{dgm}(\mathrm{Filt}(\mathbb{X}_n)))\right] \leq C\left(\frac{\log n}{n}\right)^{1/b}$$

where the constant C only depends on a and b.

Under additional technical assumptions, the corresponding lower bound can be shown (up to a logarithmic term), see Chazal et al. (2014b). By applying stability results, similar consistency results can be easily derived under alternative generative models as soon as a consistent estimator of the support under Hausdorff metric is known. For instance, from the results of Genovese et al. (2012) about Hausdorff support estimation under additive noise, it can be deduced that the minimax convergence rates for the persistence diagram estimation is faster than $(\log n)^{-1/2}$. Moreover, as soon as a stability result is available for some given representation of persistence, similar consistency results can be directly derived from the consistency for persistence diagrams.

Estimation of the persistent homology of functions. Theorem 7 opens the door to the estimation of the persistent homology of functions defined on $\mathbb{R}^d$, on a submanifold of $\mathbb{R}^d$ or more generally on a metric space. The persistent homology of regression functions has also been studied in Bubenik et al. (2010). The alternative approach of Bobrowski et al. (2014) which is based on the inclusion map between nested pairs of estimated level sets can be applied with kernel density and regression kernel estimators to estimate persistence homology of density functions and regression functions. Another direction of research on this topic concerns various versions of robust TDA. One solution is to study the persistent homology of the upper level sets of density estimators (Fasy et al., 2014b). A different approach, more closely related to the distance function, but robust to noise, consists in studying the persistent homology of the sub level sets of the distance to measure defined in Section 4.4 (Chazal et al., 2014a).

6.2 Statistic of persistent homology computed on a point cloud

For many applications, in particular when the support of the point cloud is not drawn on or close to a geometric shape, persistence diagrams can be quite complex to analyze. In particular, many topological features are closed to the diagonal. Since they correspond to topological structures that die very soon after they appear in the filtration, these points are generally considered as noise, see Figure 13 for an illustration. Confidence regions of persistence diagram are rigorous answers to the problem of distinguishing between signal and noise in these representations.

The stability results given in Section 5.7 motivate the use of the bottleneck distance to define confidence regions. However alternative distances in the spirit of Wasserstein distances can be proposed too. When estimating a persistence diagram dgm with an estimator $\widehat{\mathrm{dgm}}$, we typically look for some value η_α such that

$$P(\mathrm{d_b}(\widehat{\mathrm{dgm}}, \mathrm{dgm}) \geq \eta_\alpha) \leq \alpha,$$

for $\alpha \in (0,1)$. Let B_α be the closed ball of radius α for the bottleneck distance and centered at $\widehat{\mathrm{dgm}}$ in the space of persistence diagrams. Following Fasy et al. (2014b), we can visualize the signatures of the points belonging to this ball in various ways. One first option is to center a box of side length 2α at each point of the persistence diagram $\widehat{\mathrm{dgm}}$. An alternative solution is to visualize the confidence set by adding a band at (vertical) distance $\eta_\alpha/2$ from the diagonal (the bottleneck distance being defined for the ℓ_∞ norm), see Figure 13 for an illustration. The points outside the band are then considered as significant topological features, see Fasy et al. (2014b) for more details.

Several methods have been proposed in Fasy et al. (2014b) to estimate η_α in different frameworks. These methods mainly rely from stability results for persistence diagrams: confidence sets for diagrams can be derived from confidence sets in the sample space.

Subsampling approach. This method is based on a confidence region for the support K of the distribution of the sample in Hausdorff distance. Let $\tilde{\mathbb{X}}_b$ be a subsample of size b drawn from the sample $\tilde{\mathbb{X}}_n$, where $b = o(n/logn)$. Let $q_b(1-\alpha)$ be the quantile of of the distribution of Haus $\left(\tilde{\mathbb{X}}_b, \mathbb{X}_n\right)$. Take $\hat{\eta}_\alpha := 2\hat{q}_b(1-\alpha)$ where $\hat{q}_b$ is an estimation $q_b(1-\alpha)$ using a standard

Monte Carlo procedure. Under an (a, b) standard assumption, and for n large enough, Fasy et al. (2014b) show that

$$\begin{aligned} P\left(\mathrm{d_b}\left(\mathrm{dgm}\left(\mathrm{Filt}(K)\right), \mathrm{dgm}\left(\mathrm{Filt}(\mathbb{X}_n)\right)\right) > \hat{\eta}_\alpha\right) &\leq P\big(\mathrm{Haus}\left(K, \mathbb{X}_n\right) > \hat{\eta}_\alpha\big) \\ &\leq \alpha + O\left(\frac{b}{n}\right)^{1/4}. \end{aligned}$$

Bottleneck Bootstrap. The stability results often leads to conservative confidence sets. An alternative strategy is the bottleneck bootstrap introduced in Chazal et al. (2016b). We consider the general setting where a persistence diagram $\widehat{\mathrm{dgm}}$ is defined from the observation $(X_1, \ldots, X_n)$ in a metric space. This persistence diagram corresponds to the estimation of an underlying persistence diagram dgm, which can be related for instance to the support of the measure, or to the sublevel sets of a function related to this distribution (for instance a density function when the X_i's are in $\mathbb{R}^d$). Let $(X_1^*, \ldots, X_n^*)$ be a sample from the empirical measure defined from the observations $(X_1, \ldots, X_n)$. Let also $\widehat{\mathrm{dgm}}^*$ be the persistence diagram derived from this sample. We then can take for η_α the quantity $\hat{\eta}_\alpha$ defined by

$$P(\mathrm{d_b}(\widehat{\mathrm{dgm}}^*, \widehat{\mathrm{dgm}}) > \hat{\eta}_\alpha \,|\, X_1, \ldots, X_n) = \alpha. \tag{7}$$

Note that $\hat{\eta}_\alpha$ can be easily estimated with Monte Carlo procedures. It has been shown in Chazal et al. (2016b) that the bottleneck bootstrap is valid when computing the sublevel sets of a density estimator.

Bootstrapping Persistent Betti numbers. As already mentioned, confidence regions based on stability properties of persistence may lead to very conservative confidence regions. Based on the concepts of stabilizing statistics Penrose and Yukich (2001), asymptotic normality for persistent Betti numbers has been shown recently by Krebs and Polonik (2019); Roycraft et al. (2020), under very mild conditions on the filtration and the distribution of sample cloud. In addition, bootstrap procedures are also shown to be valid in this framework. More precisely, a smoothed bootstrap procedure together with a convenient rescaling of the point cloud seems to be a promising approach for boostrapping TDA features from point cloud data.

6.3 Statistic for a family of persistent diagrams or other representations

Up to now in this section, we were only considering statistics based on one single observed persistence diagram. We now consider a new framework where several persistence diagrams (or other representations) are available and we are interested in providing central tendency, confidence regions and hypothesis tests for topological descriptors built on this family.

6.3.1 Central tendency for persistent homology

Mean and expectations of distributions of diagrams. The space of persistence diagrams being a general metric space but not an Hilbert space, the definition of a *mean persistence diagram* is not obvious and unique. One first natural approach to define a central tendency in this context is to consider Fréchet means of distributions of diagrams. Their existence has been proved in Mileyko et al. (2011) and they have also been characterized in Turner et al. (2014a). However they are may not be unique and they turn out to be difficult to compute in practice. To partly overcome these problems, different approaches have been recently proposed based on numerical optimal transport Lacombe et al. (2018) or linear representations and kernel based methods Divol and Chazal (2020).